Backpropagation

CSC 487: Deep Learning

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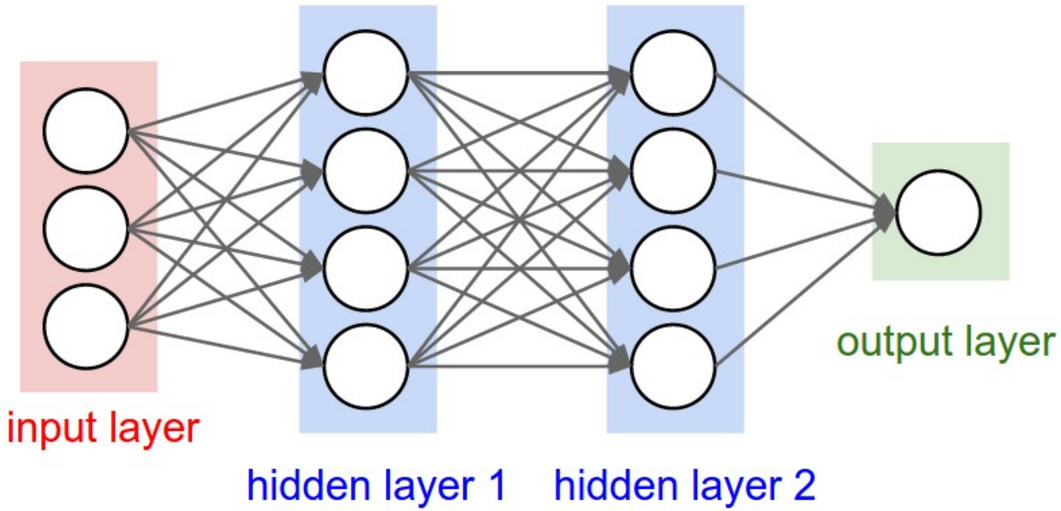
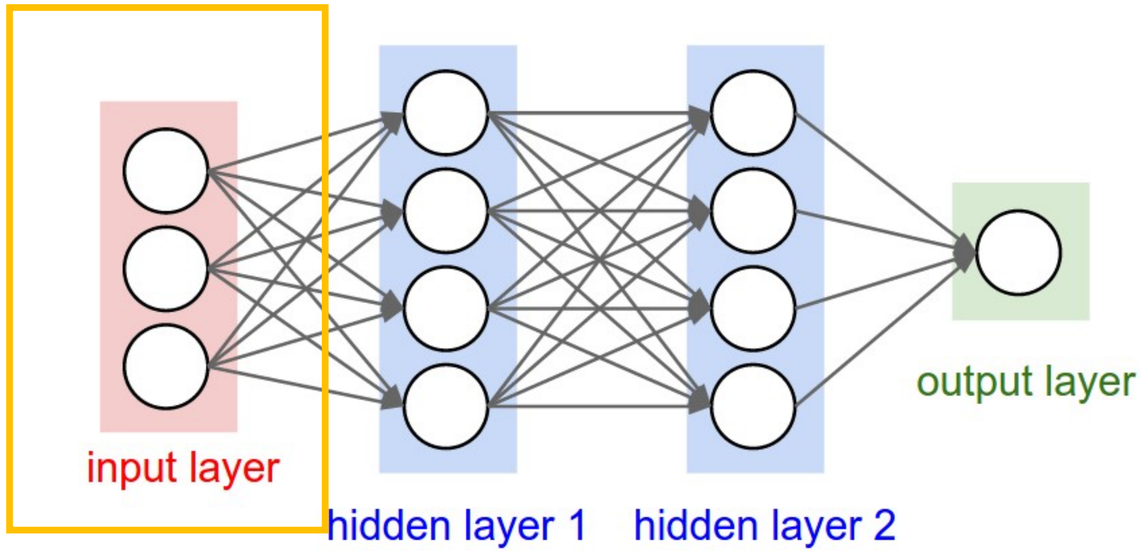
Neural network represents a non-linear function

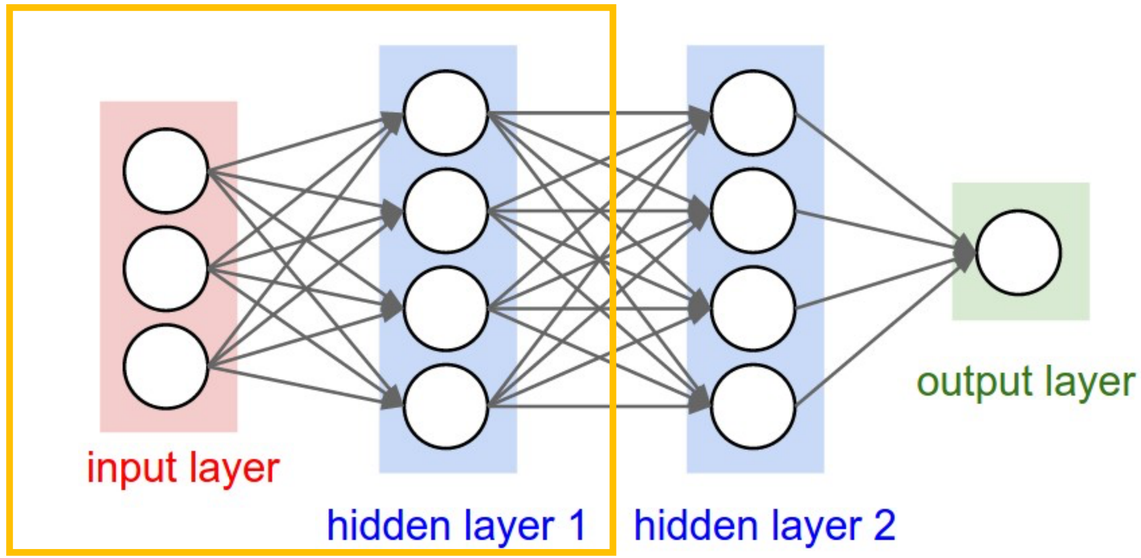


Neural network represents a non-linear function



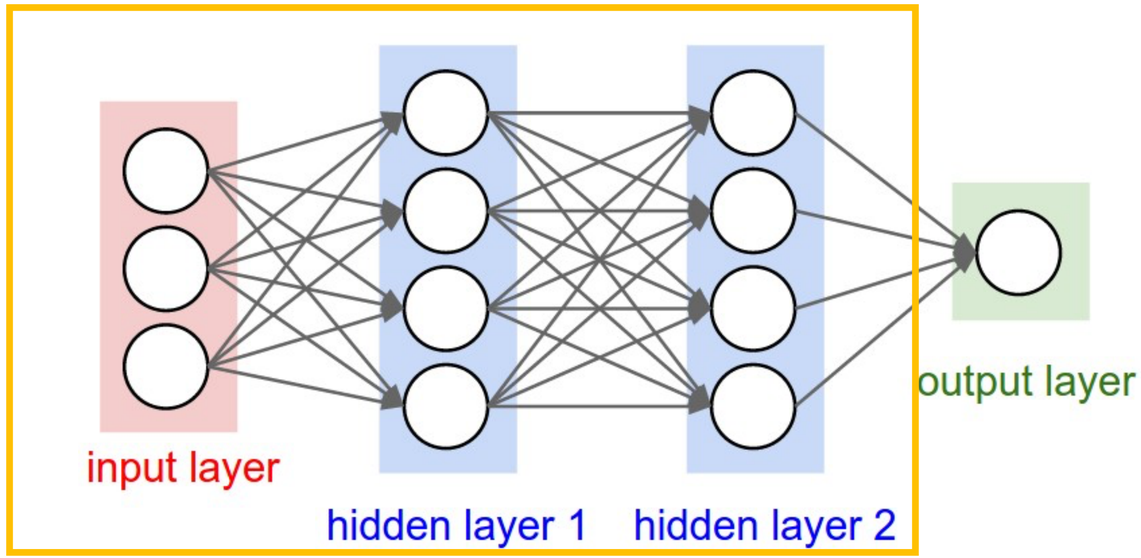
$\mathbf{x}$

Neural network represents a non-linear function



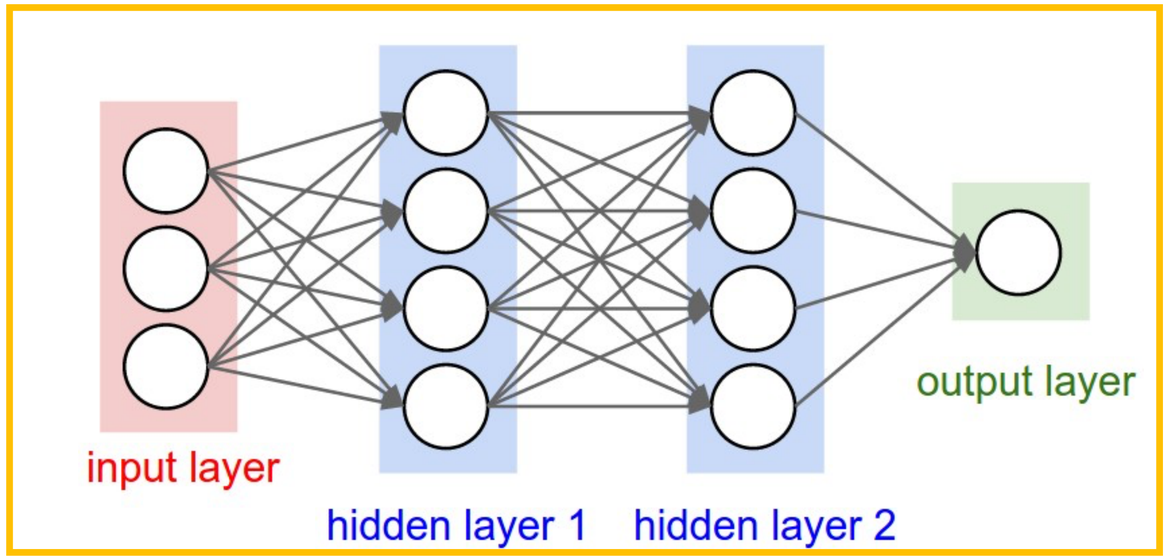
$$\sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)$$

Neural network represents a non-linear function



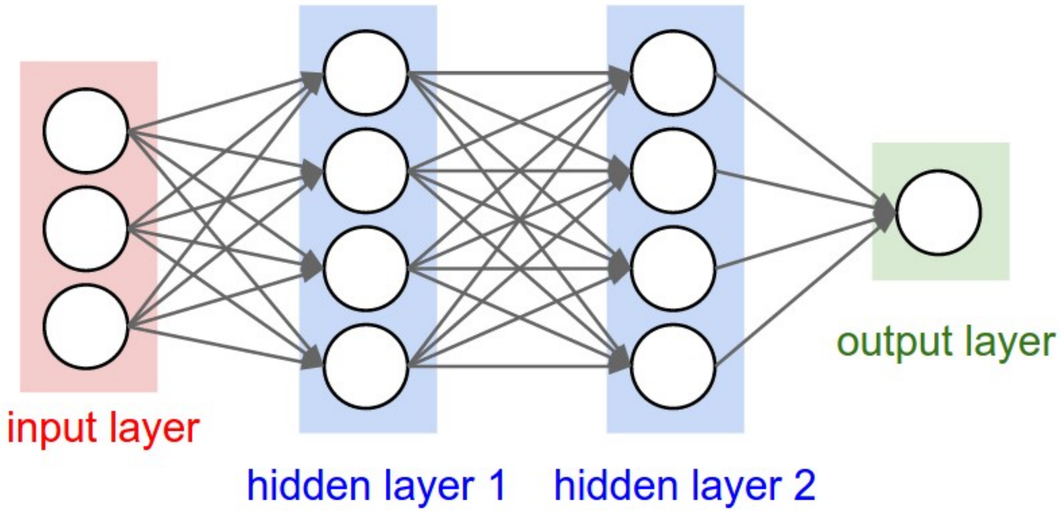
$$\sigma(\mathbf{W}_2\sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2)$$

Neural network represents a non-linear function



$$\mathbf{W}_3\sigma(\mathbf{W}_2\sigma(\mathbf{W}_1 x + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3$$

Adding a loss function



$$L = \frac{1}{2}(\mathbf{W}_3\sigma(\mathbf{W}_2\sigma(\mathbf{W}_1 x + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) - y)^2$$

Training the network

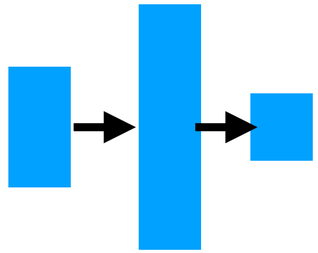
- Gradient descent:

$$w_j = w_j - s \cdot \frac{\partial L}{\partial w_j}$$

- How do we calculate $\frac{\partial L}{\partial w_j}$?

Single hidden layer example

- Assume in the remaining that we have “hidden” the bias into the weights.



$$\mathbf{W}_2\sigma(\mathbf{W}_1\mathbf{x}) = \mathbf{z}(\sigma(\mathbf{h}(\mathbf{x})))$$

$$\mathbf{h}(\mathbf{x}) = \mathbf{W}_1\mathbf{x}$$

$$\mathbf{z}(\mathbf{x}) = \mathbf{W}_2\mathbf{x}$$

Single hidden layer example

$$\frac{\partial}{\partial \mathbf{w}_1} L(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x})))) = ?$$

Remember the chain rule:

$$(f(g(x))' = f'(g(x))g'(x)$$

Generalized Chain Rule

- Suppose we have a function $L(f_1(x), f_2(x), \ldots, f_n(x))$.
- How to apply chain rule to compute $\frac{\partial L}{\partial x}$?
- The *generalized chain rule* for multivariate functions:

$$\frac{\partial L}{\partial x}(x) = \sum_{i=1}^{n} \frac{\partial L}{\partial f_i} \frac{\partial f_i}{\partial x}$$

Generalized Chain Rule

- Suppose we have a function $L(\mathbf{z}(w_j))$.
- How to apply chain rule to compute $\frac{\partial L}{\partial w_j}$?
- The *generalized chain rule* for multivariate functions:

$$\frac{\partial L}{\partial w_j}(w_j) = \frac{\partial L}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial w_j}$$

$$1 \times |\mathbf{z}| \qquad |\mathbf{z}| \times 1$$

Generalized Chain Rule

- Suppose we have a function $L(\mathbf{z}(\mathbf{w}))$.
- How to apply chain rule to compute $\frac{\partial L}{\partial \mathbf{w}}$?
- The *generalized chain rule* for multivariate functions:

$$\frac{\partial L}{\partial \mathbf{w}}(\mathbf{w}) = \frac{\partial L}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \mathbf{w}}$$

$1 \times |\mathbf{z}|$ $|\mathbf{z}| \times |\mathbf{w}|$

Single hidden layer example

- Application of chain rule for derivatives:

$$\frac{\partial L}{\partial \mathbf{W}_1} = \frac{\partial L}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \sigma} \frac{\partial \sigma}{\partial \mathbf{h}} \frac{\partial \mathbf{h}}{\partial \mathbf{W}_1}$$

Single hidden layer example

- Application of chain rule for derivatives:

$$\frac{\partial L}{\partial \mathbf{W}_1}(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x})))) = \frac{\partial L}{\partial \mathbf{z}}(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x}))))\frac{\partial \mathbf{z}}{\partial \sigma}(\sigma(\mathbf{h}(\mathbf{x})))\frac{\partial \sigma}{\partial \mathbf{h}}(\mathbf{h}(\mathbf{x}))\frac{\partial \mathbf{h}}{\partial \mathbf{W}_1}(\mathbf{x})$$

Single hidden layer example

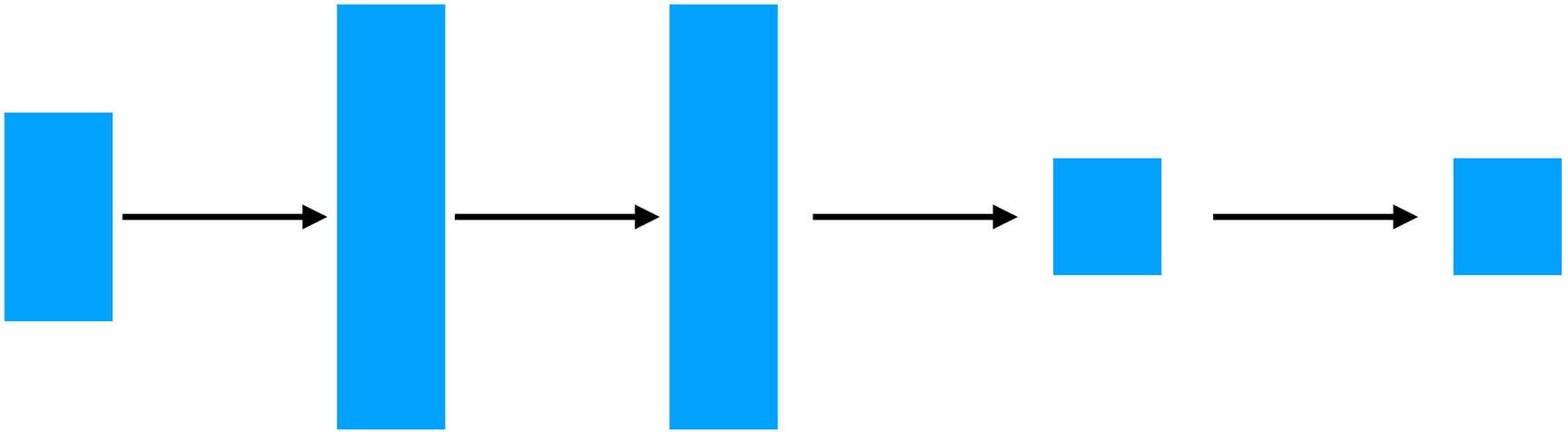
- How to compute this?

$$\frac{\partial L}{\partial \mathbf{W}_1}(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x})))) = \frac{\partial L}{\partial \mathbf{z}}(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x}))))\frac{\partial \mathbf{z}}{\partial \sigma}(\sigma(\mathbf{h}(\mathbf{x})))\frac{\partial \sigma}{\partial \mathbf{h}}(\mathbf{h}(\mathbf{x}))\frac{\partial \mathbf{h}}{\partial \mathbf{W}_1}(\mathbf{x})$$

- Could evaluate each derivative from scratch
  - Lots of redundant computation — terms are reused each time

Forward pass

Propagate input forward through network

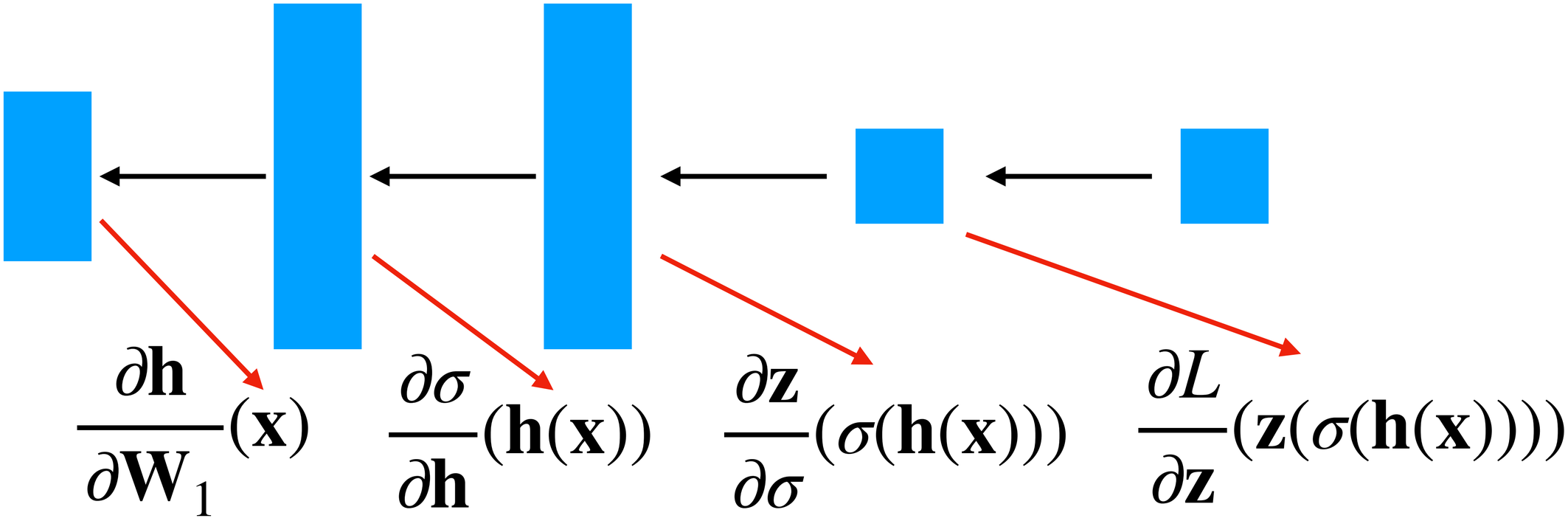


$\mathbf{x}$ $\mathbf{h}(\mathbf{x})$ $\sigma(\mathbf{h}(\mathbf{x}))$ $\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x})))$ $L(\mathbf{z}(\sigma(\mathbf{h}(\mathbf{x}))))$

Save these intermediate values!

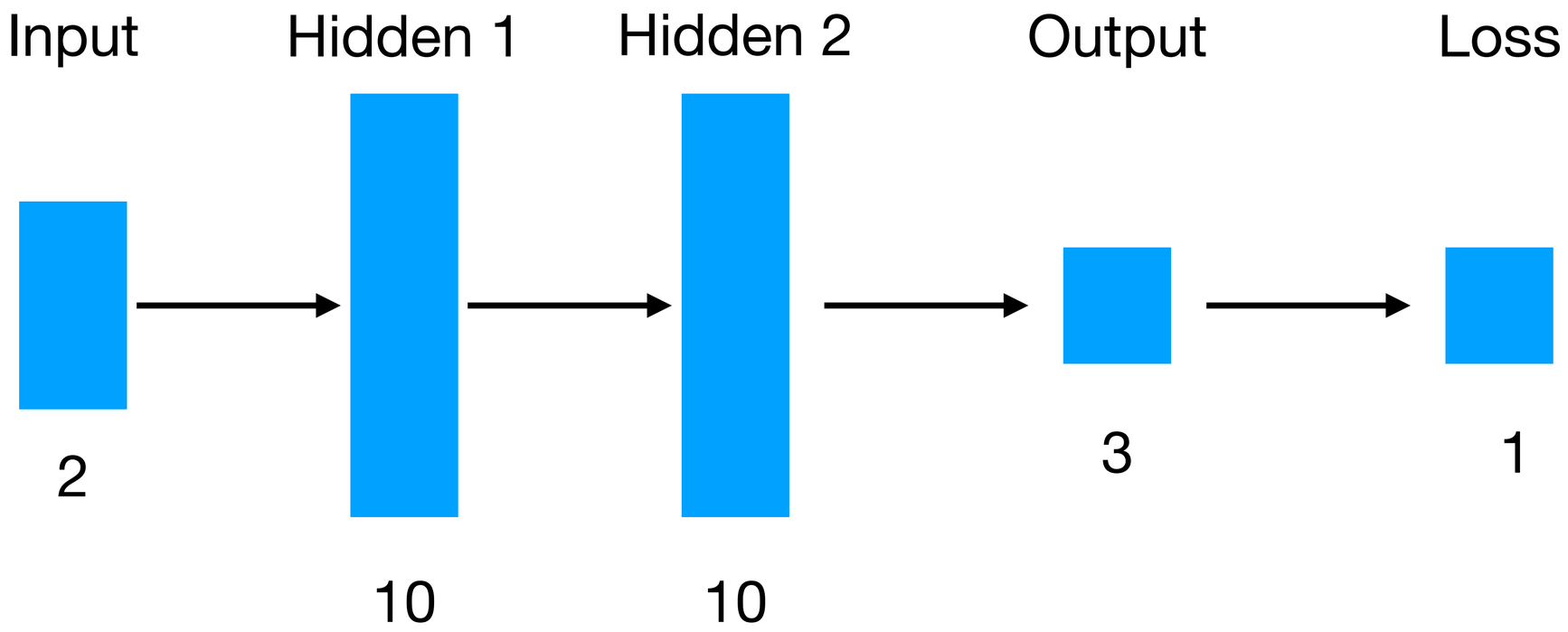
Backward pass

Propagate input forward through network



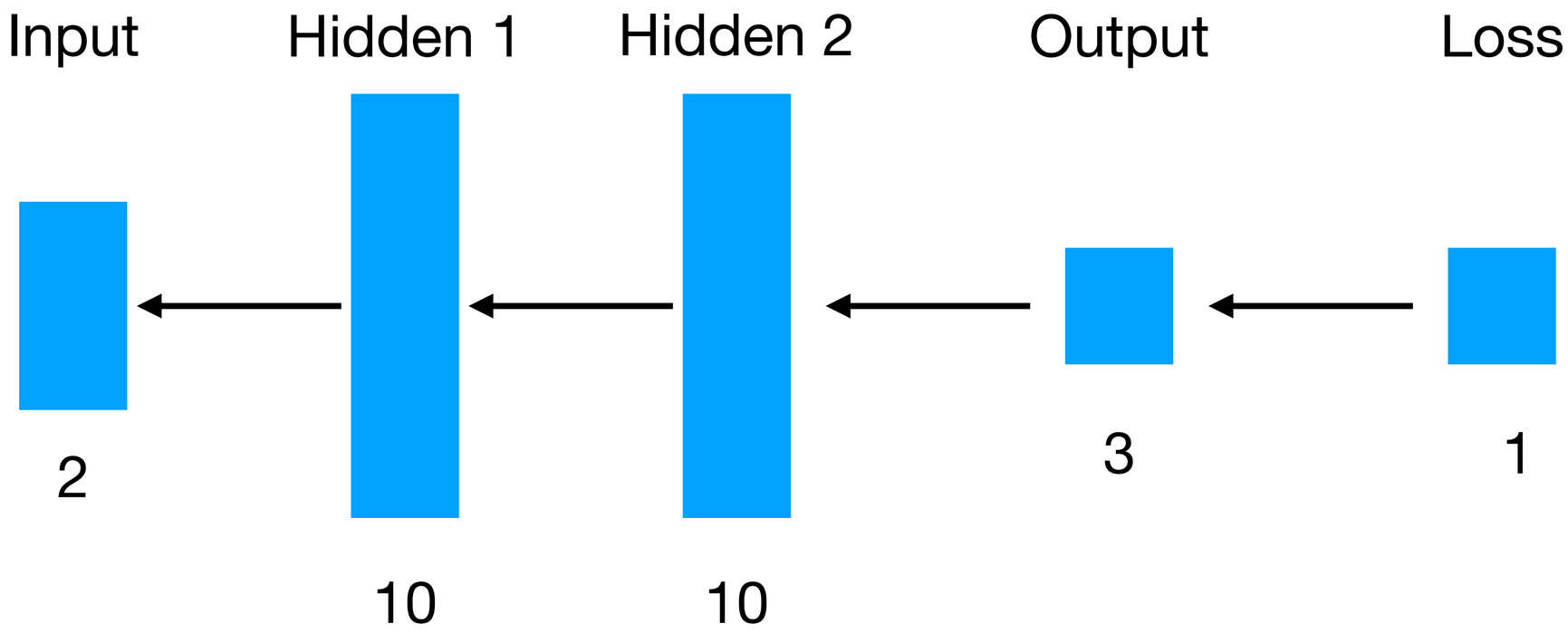
Reuse the intermediate values when computing derivatives
Update weights at each layer as error term propagates backward

Exercise



- Write down the sizes of all matrices used in the **forward** pass

Exercise



- Write down the sizes of all matrices used in the **backward** pass